A DIVERGING HISTORY CLOCK IS NOT A BLOW-UP PROOF: LEAN-CHECKED LEMMAS SHOWING WHY A SECTION 10 NAVIER–STOKES ARGUMENT DOES NOT CERTIFY FEFFERMAN (C) OR (D)

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Abstract. This is a negative-result note. It does *not* solve, and makes no claim toward, the Clay Millennium Navier–Stokes problem. An earlier argument (its "Section 10") was proposed as a certificate of finite-time blow-up, i.e. of Fefferman's statements (C) or (D). The argument used a "quadratic history clock" $\int_0^T \|\omega\|_\infty^2\,dt$ that diverges as $T \to \infty$. We explain exactly why this does not imply blow-up and machine-check the scalar real-analysis facts behind that explanation in Lean 4 with Mathlib. The facts are: (i) a pointwise bound $f \le M$ on $[0,T]$ gives $\int_0^T f \le MT$; (ii) for any f with f and f^2 integrable on $[0,T]$, $\int_0^T f \le \frac{1}{2}\big(T + \int_0^T f^2\big)$, so a quadratic history that is finite on bounded intervals can never witness divergence of the Beale–Kato–Majda (BKM) integral at a finite time; (iii) for a constant bound $M \neq 0$ the history $\int_0^T M^2$ tends to $+\infty$ as $T \to \infty$, while $\int_0^T M = MT$ for every finite T; (iv) the model profile $(T^* - t)^{-1}$ and its square are not integrable on $[0,T^*]$, which is what genuine finite-time BKM divergence looks like. There are eight theorems in all, with no `sorry`, using only Lean's standard axioms. No Navier–Stokes equation, vorticity field or BKM theorem is formalized. The BKM criterion is cited, not proved.

1. The problem statement that was attempted

Fefferman's official formulation [2] of the Navier–Stokes Millennium problem has four alternatives. (A) and (B) assert global smooth solutions on $\mathbb{R}^3$ and $\mathbb{T}^3$. (C) and (D) assert breakdown: there exist smooth, suitably decaying (resp. periodic) data, and a smooth force for (C)/(D) as stated, for which no global smooth solution exists. A proof of (C) or (D) must therefore exhibit a solution whose smoothness is lost at some *finite* time T^*. By Beale–Kato–Majda [1], if $T^* < \infty$ is maximal then

$$\int_0^{T^*} \|\omega(t)\|_{L^\infty}\,dt = \infty.$$

So a blow-up certificate must show that the BKM integral diverges on a *bounded* interval.

The Section 10 materials instead analysed a regime (2D or steady-Euler "kill switch") in which $M(t) := \|\omega(t)\|_\infty$ stays bounded, and observed that a quadratic history $H(T) = \int_0^T M^2\,dt$ grows without bound as $T \to \infty$. A request to title the repository as a resolution of (C)/(D) and to certify that argument as a zero-`sorry` proof was declined. This note records why, with the reasons machine-checked.

2. What is formally verified

Throughout, $f : \mathbb{R} \to \mathbb{R}$ is an arbitrary real function of time standing in for $t \mapsto \|\omega(t)\|_\infty$; integrals are Lebesgue interval integrals in Mathlib. File `lean/BKMCalculus.lean`, namespace `ClayCD.BKMCalculus`.

Theorem 2.1 (`bkm_integral_le_of_bound`)**.** *If $0 \le T$, f is integrable on $[0,T]$ and $f(t) \le M$ for $t \in [0,T]$, then $\int_0^T f \le MT$.*

Date: November 4, 2026. Version 2.0.0. Concept DOI (all versions): 10.5281/zenodo.23653014. Repository: `https://github.com/BenFrohman/Clay-CD-not-certified`.

Hence in the bounded kill-switch regime the BKM integral is finite on every bounded interval, and BKM gives *continuation*, the opposite of blow-up.

Lemma 2.2 (`le_half_one_add_sq`)**.** *For all real x, $x \le \frac{1}{2}(1+x^2)$.*

Theorem 2.3 (`bkm_integral_le_of_history`)**.** *If $0 \le T$ and f, f^2 are integrable on $[0,T]$, then $\int_0^T f \le \frac{1}{2}\big(T + \int_0^T f^2\big)$.*

This is the central negative result. If the history clock H is finite at every finite time, as it is in the Section 10 regime, then the BKM integral is finite at every finite time. A history clock can detect a finite-time singularity only if it is *itself* infinite on a bounded interval. Divergence "at $T = \infty$" carries no information about (C) or (D).

Theorem 2.4 (`history_diverges_at_infinity_bkm_finite`)**.** *For $M \neq 0$: $T \mapsto \int_0^T M^2\,dt$ tends to $+\infty$ as $T \to \infty$, and $\int_0^T M\,dt = MT$ for every T.*

This is the explicit counterexample to the inference "$H(T) \to \infty \Rightarrow$ blow-up". It comes from a bounded, constant M, which BKM continues globally.

Theorem 2.5 (`blowup_profile_not_integrable`, `blowup_profile_sq_not_integrable`)**.** *For $T^* > 0$, neither $t \mapsto (T^* - t)^{-1}$ nor $t \mapsto (T^* - t)^{-2}$ is integrable on $[0, T^*]$.*

This is the shape a genuine certificate would need. The second statement is derived from the first using the pointwise bound $|g| \le \frac{1}{2}(1+g^2)$, the same mechanism as Theorem 2.3.

The original two definitional identities of version 1 are retained in `lean/NotBlowup.lean` (`bounded_M_keeps_bkm_finite`, `history_can_grow_while_bkm_stays_finite`). They hold by `rfl` and are subsumed by Theorems 2.1 and 2.4, which work with actual integrals.

3. The precise gap: why this is not a Clay certificate

1. **Wrong time scale.** (C)/(D) need divergence on $[0, T^*]$ with $T^* < \infty$. The Section 10 quantity diverges only as $T \to \infty$ (Theorems 2.3, 2.4).
2. **Wrong direction.** Bounded M is exactly the hypothesis under which BKM *continues* solutions (Theorem 2.1).
3. **Wrong dimension/regime.** The 2D and steady-Euler regimes are globally regular classically, so they cannot carry a counterexample to 3D regularity.
4. **Nothing about PDE is formalized.** No equation, solution, function space or BKM theorem appears in the Lean code. The lemmas are scalar calculus. Their relevance to Navier–Stokes passes through the *cited* BKM theorem.
5. **Forced (C)/(D) claims elsewhere.** An external claim exists in `openai/NavierStokesAndEuler`. It is neither copied nor certified here.

4. Lessons and negative results

- A monotone "clock" functional that is finite at every finite time is not a blow-up witness. Theorem 2.3 turns this into a one-line test: check whether the clock can be infinite on a *bounded* interval.
- Formalizing closed forms by definition (version 1) checks nothing. Lemmas about real integrals (version 2) at least check the calculus that the argument relies on.
- An axiom audit certifies only that the stated theorems follow from standard axioms. It says nothing about whether the prose reading is the intended one.

5. Correspondence and reproducibility

Claim	Lean declaration (`ClayCD.BKMCalculus.*`)
Theorem 2.1	`bkm_integral_le_of_bound`
Lemma	`le_half_one_add_sq`
Theorem 2.3	`bkm_integral_le_of_history`
Theorem 2.4	`history_diverges_at_infinity_bkm_finite`
Theorem 2.5	`blowup_profile_not_integrable`, `blowup_profile_sq_not_integrable`

Toolchain `leanprover/lean4:v4.30.0-rc1`, Mathlib `v4.30.0-rc1` (commit `0c154d67`). To reproduce: `cd lean && lake exe cache get && lake build && lake env lean AxiomAudit.lean`. CI rebuilds on every push, fails on `sorryAx`, on any nonstandard axiom, or on any `sorry`/`admit`/`native_decide`/`axiom` in source, and builds this PDF.

Axiom audit. Verbatim `#print axioms` output: every one of the 8 theorems depends on exactly `[propext, Classical.choice, Quot.sound]`.

Declaration	Axioms
`ClayCD.NotBlowup.bounded_M_keeps_bkm_finite`	standard three
`ClayCD.NotBlowup.history_can_grow_while_bkm_stays_finite`	standard three
`ClayCD.BKMCalculus.bkm_integral_le_of_bound`	standard three
`ClayCD.BKMCalculus.le_half_one_add_sq`	standard three
`ClayCD.BKMCalculus.bkm_integral_le_of_history`	standard three
`ClayCD.BKMCalculus.history_diverges_at_infinity_bkm_finite`	standard three
`ClayCD.BKMCalculus.blowup_profile_not_integrable`	standard three
`ClayCD.BKMCalculus.blowup_profile_sq_not_integrable`	standard three

Nothing here has been peer reviewed.

References

[1] J. T. Beale, T. Kato, A. Majda, *Remarks on the breakdown of smooth solutions for the 3-D Euler equations*, Comm. Math. Phys. **94** (1984), 61–66.

[2] C. L. Fefferman, *Existence and smoothness of the Navier–Stokes equation*, in: The Millennium Prize Problems, Clay Math. Inst. (2006), 57–67.

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